

T2-1

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad AX = \begin{bmatrix} a_1x_1 + b_1x_2 + c_1x_3 \\ \vdots \\ a_4x_1 + b_4x_2 + c_4x_3 \end{bmatrix} = x_1 \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} + x_2 \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} + x_3 \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

$AX =$ linear combination of col vectors of A with entries of x as wts.

T2-4 Every lin map $L: \mathbb{R}^c \rightarrow \mathbb{R}^r$ is of form $L(x) = AX$ for a unique $r \times c$ matrix A .

Note. $X = \begin{bmatrix} x_1 \\ \vdots \\ x_c \end{bmatrix} = x_1 e_1 + \dots + x_c e_c$ where $e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ in i th slot

$\Rightarrow$ We must have $L(x) = x_1 L(e_1) + \dots + x_c L(e_c) \stackrel{\#1}{=} \begin{bmatrix} L(e_1) & \dots & L(e_c) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_c \end{bmatrix}$

sp. case $c=1$ $L: \mathbb{R} \rightarrow \mathbb{R}^r$
 Note. $\vec{x} = x \cdot \vec{T}$

$L(\vec{x}) = x L(\vec{T})$ i.e. If $L(\vec{T}) = \begin{bmatrix} v_1 \\ \vdots \\ v_r \end{bmatrix}$
 $L(\vec{x}) = \begin{bmatrix} v_1 \\ \vdots \\ v_r \end{bmatrix} [x]$

linear map defⁿ: f such that $f(v_1 + v_2) = f(v_1) + f(v_2)$, $f(cv) = c f(v)$

Claim Linear maps $\mathbb{R}^c \rightarrow \mathbb{R}^r$ $\xleftrightarrow{\text{bijection}}$ $r \times c$ matrices

$$f(x) = Ax$$

 A f matrix with cols $f(e_1) \dots f(e_c)$

Imp. subspaces associated to linear f associated to matrix A

check that these are subspaces

① $\text{Nul}(A) = \ker(f) := \{x \mid f(x) = Ax = 0\}$ subspace of domain of f .

② $\text{Col}(A) = \text{Im}(f) := \{f(x) = Ax \mid x \in \text{domain}\}$ subspace of codomain of f .
= Span of cols of A .

Defⁿ A basis for a vector space W is a subset B of W such that each $w \in W$ can be written uniquely as a linear combination of vectors from B .

Facts: Any v.s. space W has a basis
Any two bases of W have the same cardinality called the dimension of W

Proofs: Later (for "finite dimensional" W)
defⁿ?

Let's assume for now that

we have a finite ordered basis $w_1, \dots, w_n$ for W .

Then any vector w can be written uniquely

as $w = c_1 w_1 + \dots + c_n w_n$. We call $\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ the coordinates of w wrt. B .

Failure of uniqueness would mean we have $\begin{cases} a_1 w_1 + \dots + a_n w_n = b_1 w_1 + \dots + b_n w_n \\ \text{with } a_i \neq b_i \text{ for some } i. \end{cases}$

Def A set S of vectors in W is called linearly indep if

$$\Leftrightarrow \sum_i c_i w_i = 0 \text{ with some } c_i \neq 0$$

$$\left(\sum_{v_i \in S} c_i v_i = 0 \Leftrightarrow \text{each } c_i = 0 \right)$$

(finite sum)

Observe/exercise: A basis for W is a linearly independent set that spans W

Ex A minimal spanning set is a basis

A maximal lin. indep set is a basis

Here "maximal/minimal" is with respect to set containment